Statistical Modelling & Analysis (STAT2004)
Applied Class (Week 5)

We will go through Questions 3 & 4 together in the Applied Class. Try the other questions yourself during the week and discuss your results with your tutor next week.

1. Suppose that $Y \sim \mathsf{N}(\mathbf{X}\boldsymbol{\beta}, \sigma^2 I_n)$, where $\mathbf{X}$ is an $n \times p$ matrix, $n > p$ and $\sigma^2 > 0$. If $\mathbf{X}$ has full column rank, (that is, there is no $\mathbf{c} \in \mathbb{R}^p \setminus \{\mathbf{0}\}$ such that $\mathbf{X}\mathbf{c} = \mathbf{0}$) the MLE is unique and given by
$$\hat{\boldsymbol{\beta}} = (\mathbf{X}^T\mathbf{X})^{-1}\mathbf{X}^T Y.$$
   (a) Show that $\mathbf{X}^T\mathbf{X}$ is invertible if and only if $\mathbf{X}$ has full column rank.
   (b) Let $\mathbf{c} \in \mathbb{R}^p$. An unbiased estimator of $\mathbf{c}^T\beta$ is $\mathbf{c}^T\hat{\beta}$. Is there a vector $\mathbf{c} \in \mathbb{R}^p \setminus \{\mathbf{0}\}$ for which $\mathrm{Var}(\mathbf{c}^T\hat{\beta}) = 0$?

2. Consider the regression-through-the-origin model
$$Y_i = \beta x_i + \epsilon_i,$$
   where the ϵ_i are IID $\mathsf{N}(0, \sigma^2)$ and $x_1, x_2, \ldots$ is a fixed (ie non-random) sequence. Assume that $x_1 \neq 0$.
   (a) Let $\hat{\beta}_n$ denote the MLE of β from the first n observations. Write down an expression for $\hat{\beta}_n$.
   (b) State the distribution of $\hat{\beta}_n$.
   (c) Give a necessary and sufficient condition on $x_1, x_2, \ldots$ for $\hat{\beta}_n$ to be a consistent estimator of β, that is for all $\delta > 0$,
$$\lim_{n\to\infty} \mathbb{P}\left(|\hat{\beta}_n - \beta| > \delta\right) = 0$$
   (d) Interpret the condition in part (c) in terms of the sensitivity of the mean response to changes in β. Explain why increasing the number of observations does not necessarily guarantee consistency.
   (e) Suppose $x_j \neq 0$ for all j. An alternative estimator of β is
$$\tilde{\beta}_n = n^{-1}\sum_{j=1}^{n} \frac{Y_j}{x_j}.$$
   Use the Cauchy–Schwarz inequality $\left(\sum_{j=1}^n a_j b_j\right)^2 \leq \left(\sum_{j=1}^n a_j^2\right)\left(\sum_{j=1}^n b_j^2\right)$ to show $\mathrm{MSE}(\hat{\beta}_n) \leq \mathrm{MSE}(\tilde{\beta}_n)$. When is the MSE of the two estimators equal?

3. Suppose that the true regression model is
$$Y = \mathbf{X}\boldsymbol{\beta} + \gamma\mathbf{z} + \boldsymbol{\epsilon}, \qquad \boldsymbol{\epsilon} \sim \mathsf{N}_n(\mathbf{0}, \sigma^2\mathbf{I}_n),$$
   where $\mathbf{X}$ is an $n \times p$ matrix with full column rank and $\mathbf{z} \in \mathbb{R}^n$ contains the observed values of an additional explanatory variable.
   Suppose that $\mathbf{z}$ is omitted and the model containing only the explanatory variables in $\mathbf{X}$ is fitted. The resulting estimator is
$$\hat{\boldsymbol{\beta}} = (\mathbf{X}^T\mathbf{X})^{-1}\mathbf{X}^T Y.$$
   (a) Find $\mathbb{E}(\hat{\boldsymbol{\beta}})$ under the true model.
   (b) Suppose that $\mathbf{z}$ is orthogonal to every column of $\mathbf{X}$. Show that $\hat{\boldsymbol{\beta}}$ is an unbiased estimator of $\boldsymbol{\beta}$.
   (c) Suppose that the fitted model contains an intercept. What does the condition that $\mathbf{z}$ is orthogonal to the intercept column imply about the observed values in $\mathbf{z}$?

(d) Let

$$\mathbf{H} = \mathbf{X}(\mathbf{X}^T\mathbf{X})^{-1}\mathbf{X}^T$$

be the influence matrix for the fitted model, and let

$$\mathbf{e} = Y - \mathbf{X}\hat{\boldsymbol{\beta}}$$

be the residual vector. Assume that $\mathbf{z}$ is orthogonal to every column of $\mathbf{X}$.

i. Show that $\mathbf{Hz} = \mathbf{0}$.

ii. Find an expression for $\mathbf{e}$.

iii. Find $\mathbb{E}(\mathbf{e})$ and $\mathrm{Var}(\mathbf{e})$.

iv. Explain what these results show about the effect of omitting $\mathbf{z}$.

4. The `sat` data in the `faraway` package contain state-level educational expenditure and SAT results for the 50 US states in 1994–95. Fit the model

$$\texttt{total} = \beta_0 + \beta_1\texttt{expend} + \beta_2\texttt{salary} + \beta_3\texttt{ratio} + \beta_4\texttt{takers} + \epsilon.$$

Use appropriate diagnostic quantities and plots to answer the following questions. For every plot displayed, explain what feature is relevant and what conclusion you draw. Do not display a plot unless you have something substantive to say about it.

```
library(faraway)
?sat
sat_lm <- lm(total ~ expend + salary + ratio + takers, data = sat)
```

(a) Which state has the largest Cook's distance? Report its Cook's distance, leverage and standardised residual. Explain why this state is influential.

(b) How do the estimated coefficients change when the observation for Utah is removed from the dataset? Which coefficients change most? Do the principal conclusions from the model change?

(c) West Virginia has the largest absolute standardised residual. Why does deleting West Virginia have less effect on the estimated coefficients than deleting Utah?

(d) California has a relatively high leverage of 0.282. Does deleting California have a large effect on the estimated coefficients? Explain your answer using both its leverage and its standardised residual.